

UNIT – 4
ASSESSMENT – 3
OUTPUTS

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1.

```
Episode 1

State: High_Risk | Action: Medication | Reward: 4 | Next: Low_Risk | Q: 0.4
State: Low_Risk | Action: Diet | Reward: 0 | Next: Low_Risk | Q: 0.0
State: Low_Risk | Action: Exercise | Reward: 0 | Next: Low_Risk | Q: 0.0

Episode 2

State: High_Risk | Action: Medication | Reward: 4 | Next: Low_Risk | Q: 0.796
State: Low_Risk | Action: Monitor | Reward: -1 | Next: Moderate_Risk | Q: -0.1
State: Moderate_Risk | Action: Exercise | Reward: 1 | Next: Low_Risk | Q: 0.1

...

FINAL Q TABLE

      Diet  Exercise  Medication  Monitor
Low_Risk      ...      ...      ...      ...
Moderate_Risk  ...      ...      ...      ...
High_Risk      ...      ...      ...      ...
Critical_Risk  ...      ...      ...      ...

FINAL LEARNED POLICY

Low_Risk -> ...
Moderate_Risk -> Exercise
High_Risk -> Medication
```

2.

FIRST 5 ROWS

```
-----  
    sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)  target  
0           5.1         3.5           1.4           0.2         0  
1           4.9         3.0           1.4           0.2         0  
2           4.7         3.2           1.3           0.2         0  
3           4.6         3.1           1.5           0.2         0  
4           5.0         3.6           1.4           0.2         0
```

DATA TYPES

```
-----  
sepal length (cm)    float64  
sepal width (cm)     float64  
petal length (cm)    float64  
petal width (cm)     float64  
target               int64
```

MISSING VALUES

```
-----  
sepal length (cm)    0  
sepal width (cm)     0  
petal length (cm)    0  
petal width (cm)     0  
target               0
```

Q-LEARNING PARAMETERS

```
-----  
Learning Rate (alpha) : 0.1  
Discount Factor (gamma): 0.9  
Exploration Rate (epsilon): 0.2  
Episodes: 100
```

Q-TABLE EVOLUTION

```
-----  
  
Episode 1  
State 0: [0. 0. 0. 0.]  
State 5: [0. 0. 0. 0.]  
Actions: ['UP', 'DOWN', 'LEFT', 'RIGHT']
```

```
Episode 50  
State 0: [...]  
State 5: [...]  
Actions: ['UP', 'DOWN', 'LEFT', 'RIGHT']
```

```
Episode 100  
State 0: [...]  
State 5: [...]  
Actions: ['UP', 'DOWN', 'LEFT', 'RIGHT']
```